

Project Proposal: Empirical Analysis of Minimum Spanning Tree Algorithms

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Project Description

The goal of this project is to perform an empirical study and implementation of Minimum Spanning Tree algorithms. We will focus on comparing Prim's Algorithm and Kruskal's Algorithm by simulating a state-level infrastructure project. Our scenario involves connecting major cities with fiber optic cables along existing highway routes to achieve the minimum total distance.

Data & Algorithms

Following the project requirements, we will utilize METAL graph data and supplemental information from the Travel Mapping project. We plan to focus on regions such as New York State, treating highway intersections as vertices and road segments as edges. We will analyze the efficiency of our chosen algorithms as the number of vertices and edges increases, mirroring the methodology of our previous empirical studies on sorting algorithms.

Major Milestones & Schedule

We have established the following milestones to ensure the project is feasible for the May 1 Academic Showcase:

- **Week 1:** Begin the implementation of Prim's algorithm and Kruskal's algorithm
- **Week 2:** Finish implementation of algorithms and start empirical analysis
- **Week 3:** Finish empirical analysis and start planning for academic showcase
- **Week 4:** Finish preparations for academic showcase
- **Week 5:** Academic showcase and submission